



Validating eCockpit Systems: Challenges and Way Forward

Abstract:

With the shift towards a digital lifestyle, automotive cockpit systems are enhancing their integrated capabilities. Infotainment units are merging with a digital instrument cluster and various cockpit displays. The integration of these elements leads to an intuitive, easy to use HMI. This development is being called an Electronic Cockpit (eCockpit) system or an Integrated Cockpit system. Doing so reduces material costs for an OEM by consolidating multiple functions onto a single ECU without any compromise in safety and functionality. It also provides means of separating mixed critical systems such as infotainment and cluster functions. This white paper explores the various challenges in automating the validation of such eCockpit systems and a proposed solution for a unique scalable and easy to maintain framework to address these challenges.

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Introduction



The [automotive cockpit systems market](#) is expected to register a CAGR of 8.66% by 2025. The main factor driving its growth is the heightened demand for mobility services inside the vehicle. This demand is mostly expected to come from mobility enablers such as ridesharing and taxi service providers. The other component for this surge in demand is in the luxury car segment post-pandemic especially in emerging markets such as China, Brazil, and India.

As we move towards a digital lifestyle, automotive cockpit systems are becoming more integrated. The center console also known the infotainment unit is merging with a digital instrument cluster and all other cockpit displays such as the heads-up display, passenger display, rear-seat entertainment displays, and side-view mirror displays. The integration of these elements will lead to an intuitive, easy to use HMI. This development is being called an Electronic Cockpit (eCockpit) system or an Integrated Cockpit system. Traditionally, these display systems have been housed as separate Electronic Control Units (ECUs) each powered by a high-end processor. An eCockpit system helps reduce material costs for an OEM by consolidating multiple functions onto a single ECU without any compromise in safety and functionality. Such systems also provide means of separating mixed critical systems such as infotainment and cluster functions.

Android as an operating system is rapidly gaining market share for cockpit platforms that were once dominated by QNX. This is happening primarily due to two shifts in the automotive industry i.e., connectivity and electrification. The ability to stay connected to the cloud presents an opportunity for OEMs to unlock a more personalized, data-driven experience for the customer, while electrification calls for a change in the entire vehicle's architecture.

Android being the operating system with the highest market share in handhelds, has automatically become a platform of choice for the cockpit to unify the digital experience inside and outside the vehicle. This coupled with the growth and



penetration of the smartphone and in-vehicle connectivity options indicates that the market for automotive cockpit solutions is at an inflection point.

All these drivers have put the validation aspect in the development of the cockpit system back into focus. With the increase in the number of features and the penetration of the digital lifestyle into the eCockpit units, most automotive OEMs and Tier-1 suppliers are facing many challenges in trying to test and validate the systems effectively.

The amount of software used to control the system is increasing due to the rate at which the software is being updated and enhanced. A premium class vehicle today consists of more than 100 million lines of code; a number far higher than that of an F-35 fighter jet, with just 24 million lines of code. As per forecasts, [software content in a vehicle is going to increase to about 750 million lines of code by 2025](#), paving the way for electric and autonomous mobility.

With the base software delivered from one team, the middleware from another team, and the application developed by OEMs, the integration of these different software releases and validating them as one is a big challenge. Not all releases are delivered at once, hence the frequency of testing increases with each delivery. This coupled with concepts such as Continuous Integration and DevOps provides the need to test

the software as soon as changes are made via an automated validation framework to handle these load types. The need to automate the series of regression tests at many points throughout development is evident in all companies designing these systems.

The market size for validation looks to be increasing owing to many factors as discussed earlier. Any investment for the development of a new system will have to proportionately invest significantly for system validation. This is the main reason why the trend for the validation market is growing yearly as seen below.

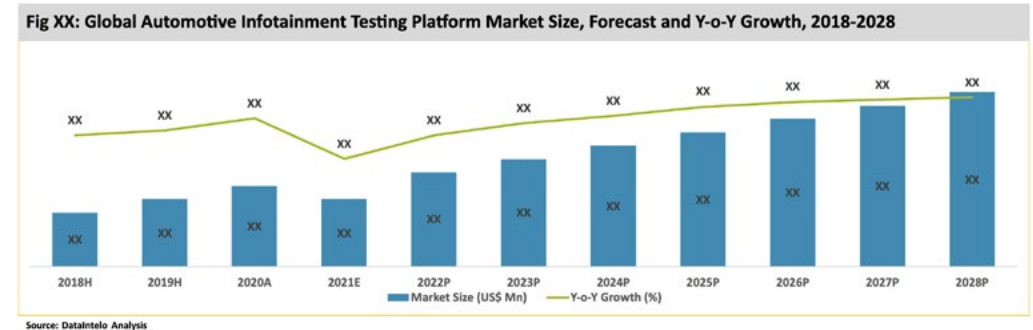


Figure 1: Global Automotive Infotainment Testing Platform Market Size Predictions | [Source](#)



Challenges In Automated Validation Of eCockpit System



One major factor that hinders validation is the availability of hardware and the limitations of both hardware and software toolchains since it is designed for one standard at a time. Therefore, validation engineers are investigating how they can get maximum reuse out of their testing platform by using flexible tools to test multiple pieces within the same infrastructure.

Some of the other factors which pose a challenge in automated validation of an eCockpit are:

- **Prioritization:** With increasingly complex features and hardware added into the system, the number of test cases to be executed is increasing exponentially. Automating these new scenarios overloads to the automated test system which slows down the entire validation cycle.
- **Communication within the cross-functional teams:** Automating new features into the existing automation framework requires the automation team to be in contact with many other stakeholders other than the manual validation team. This includes developers, business analysts, and technical



architects to confirm if features are working before they can start the automating process. For example, when automating a hands-free calling feature in an eCockpit, the middleware code is delivered from the Bluetooth stack supplier, the board support comes from the hardware vendor, and the application and HMI user interfaces comes from the internal software team. The tester must connect with all these team SMEs to understand the capabilities and specifications of the features before commencing automation. Hence, collaborating with all these stakeholders is a challenge in terms of time and effort.

- **Picking the right tool:** Based on the nature and extent of the testing to be done, the toolchain is selected. Budget availability for procuring the required infrastructure to support the toolchain is also a key factor. There are plenty of tool chain options for each use case such as HMI automation, test case creation etc., This also includes testing hardware components such as USB, audio, and video. With that said, when it comes to developing an automation suite, tool chains need be evaluated and selected for easy integration with the existing solution. There are few tool chains in the market that provides end-to-end support for both hardware and software component testing. These decisions hence require not only budgetary availability but also compatibility, scalability, and ease in performing operations.

- **Upfront capital expenditure investment:** The initial phase of implementing automation is the most expensive part. There are also many hidden costs to the solution such as the framework, libraries, or external functions which need to be procured for the complete working of the system.
- **High maintenance:** The traditional script-based testing needs frequent updates to keep up with highly dynamic accelerated software release process. This results in an overwhelming amount of test automation false positives that require high maintenance to address.



Challenges in Testing Lifecycle Of Automation & Verification of an eCockpit System



Integration testing

Integration testing is defined as testing performed to expose defects in interfaces and the interactions between integrated components or systems as per International Software Testing Qualifications Board (ISTQB®). Integration tests can be manual or automated. However, with eCockpit systems integration tests are becoming more standalone and software-based. This is caused by the interfaces changing from being hardware driven ones such as CAN to software interfaces. It is also challenging to automate these tests as multiple systems need to be validated concurrently. One such example could be the validating of HVAC, cluster, and IVI displays all at once for turn-by-turn navigation use cases at the software level without injecting CAN messages.

Performance testing

Performance testing is done primarily to determine the performance of a component or system. With multiple displays in an eCockpit, there comes the difficulty in validating the KPI requirements across displays. Taking the same example as previously discussed, validating of turn-by-turn navigation or displaying an incoming call notification in all the three screens at virtually the same time may come with a latency of 100ms.



Delivering a premium and immersive audio/video experience in eCockpit means ensuring robustness, quality, and safety of the content delivered to the end-users. Validating this in an eCockpit becomes even more challenging due to the number of input sources, and difficulty in automating tests for media and acoustics quality.

Automating the entire flow of the screen transitions or a complete flow of a feature also requires cumbersome computation to detect errors. Image-based tests are not real-time and can be found only in post-processing time. The challenge increases multifold when it comes to eCockpit with multiple screens and multi-modal interfaces.

Functional Testing

When the number of components and feature sets of a system increases, the complexity to analyze the functional specification also increases. For example, automated tuning and testing of voice recognition needs special considerations such as anechoic assessment, wake word & command recognition, false alarm rate, and barge-in.

Similarly, while automating the navigation feature testing, one needs to verify the voice and video frames simultaneously. Interoperability testing also becomes a challenge when one needs to test for multiple interfaces, operating systems, and smartphone

variants. Moreover, while certifying for a particular technology like CarPlay and Bluetooth, meeting all specifications becomes difficult with multiple subcomponents.

Regression Testing

While developing an eCockpit, manufacturers get software releases from multiple vendors. When the frequency of software integration and releases is high, it becomes difficult to execute change-related testing to detect whether defects have been introduced or uncovered in unchanged areas of the software. That is when it becomes necessary to think of an alternative automation approach to create the regression suite before each release.

Test Coverage

Due to the ever-increasing number of features and use cases along with end-customer requirements for eCockpit, the testing scenarios and requirements are increasing drastically. Achieving test coverage for a software-defined car necessitates a rethink of the test automation approach to leapfrog into the next level of testing.



The Way Forward



It's evident that additional complex features are being added to the eCockpit and the amount of data collected during the testing cycle is huge. Therefore, only relying on manual testing is not an option. Automation is the way to go forward as it provides a solution considering time and cost factors involved in addressing go-to-market needs. Therefore, test automation is a good choice but it does bring a few challenges.

There are other factors also which are pointing towards automated testing such as continuous delivery, distributed and complex application architecture, and projects with Agile and DevOps philosophies engrained in them. With more complex features added to the system, AI and ML technology are also key drivers towards automated testing. Regardless of the testing strategy chosen and the toolchains selected, there is always the concern of scaling the testing systems. To do so, we need to focus more on the design and organization of tests and the correct usage of the toolchain.



In conclusion, it is found that going forward validation systems will become more complex and highly cumbersome. To tackle this situation, moving towards an intelligent automated system seems to be the only way at this point in time. The use of latest technologies such as augmented reality, virtual reality, artificial intelligence, and machine learning could help reduce the complexity of tasks and improve the efficiency of the output. If we pursue the traditional method of testing strategies, within a few years it would become outdated and highly difficult to maintain and scale.

The proposed solution takes all the above points into consideration and provides a scalable, maintainable, and useable approach to address the current problem. The framework is flexible to accommodate any automation toolchain and integrates the solution to the cloud server for remote management. It also provides automated test case preparation and automated regression test suite creation from either a structured or unstructured data source. Post the test execution, the framework also provides features to analyze the logs, bug reports, test metrics, and vital system parameters to provide an intelligent report. This aids in rapid root cause analysis from the huge data logs and also provides features such as power testing, load testing, and performance testing to make sure the system is stable across all working environments.

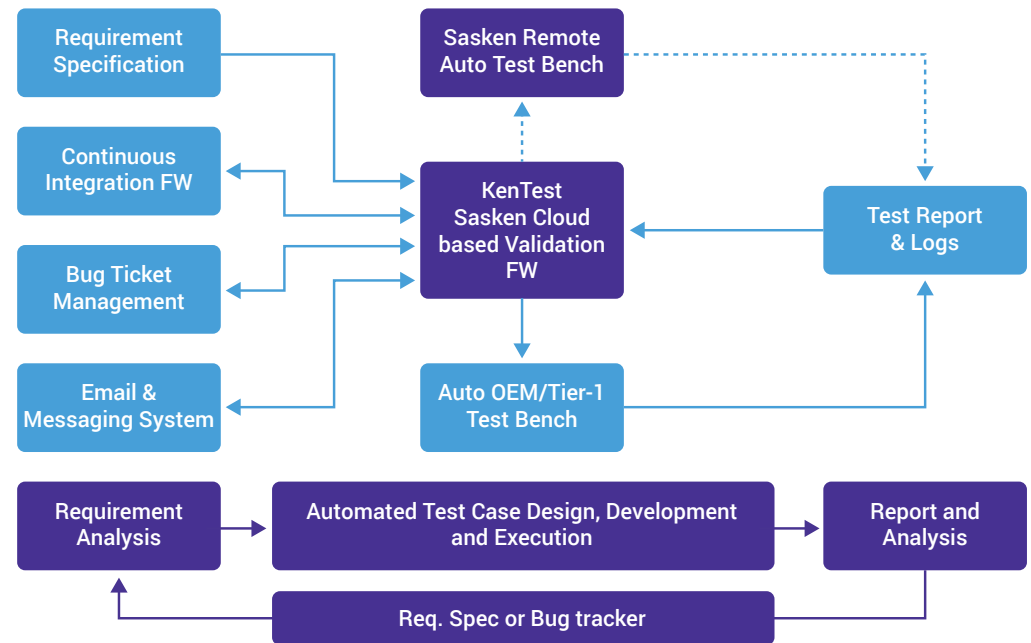


Figure 2: Sasken's Scalable Framework for Effective Testing



Key Takeaways



- The extent to which a digital lifestyle has been introduced to vehicle is high and it looks to keep growing
- More and more complex features are added which require multiple scenario-based validations which are inefficient when using manual validation
- Repetitive tasks that are prone to inattentive errors & tasks with high reuse across workflows are ideal candidates for automation
- Creating good quality test data is ideal for getting the maximum out of automated systems
- The use of machine learning in test suite creation and data analytics in test reporting improves the quality of the feature/product
- Technologies such as virtual reality can help in testing the system even before the actual hardware and software are ready



About The Authors

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About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 30 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.





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